

Module 9: Linear Programming — Answer Key

Prequiz

1. Labor:

$$3x + 2y \leq 120.$$

Land:

$$x + y \leq 50.$$

Nonnegativity: $x \geq 0, y \geq 0$. Objective:

$$\text{maximize } P = 200x + 150y.$$

2.

$$f(0, 0) = 0, \quad f(8, 0) = 40, \quad f(4, 6) = 68, \quad f(0, 8) = 64.$$

Maximum = $\boxed{68}$ at $\boxed{(4, 6)}$.

3.

$$C(8, 0) = 24, \quad C(2, 6) = 36, \quad C(0, 10) = 50.$$

Minimum = $\boxed{24}$ at $\boxed{(8, 0)}$.

4.

$$f(0, 0) = 0, \quad f(4, 0) = 12, \quad f(3, 2) = 15, \quad f(0, 5) = 15.$$

Optimal vertices: $\boxed{(3, 2)}$ and $\boxed{(0, 5)}$. Every point on the edge joining them is also optimal.

5. Check nearby feasible **integer** points, because the continuous optimum uses fractional servings and is not directly allowable when servings must be whole.

Matching quiz

1. Bread:

$$2x + y \leq 80.$$

Filling:

$$3x + 4y \leq 150.$$

Also $x, y \geq 0$. Objective:

$$\text{maximize } R = 4x + 5y.$$

2.

$$R(0, 0) = 0, \quad R(6, 0) = 36, \quad R(4, 5) = 59, \quad R(0, 7) = 49.$$

Maximum = $\boxed{59}$ at $\boxed{(4, 5)}$.

3. For $C = 5x + 2y$:

$$C(6, 0) = 30, \quad C(3, 6) = 27, \quad C(0, 12) = 24.$$

Minimum = $\boxed{24}$ at $\boxed{(0, 12)}$.

4. Every point on the edge joining the two tied adjacent vertices is optimal.

5. 7.5 machines is not physically possible if machines must be whole. Check nearby feasible integer points, such as points with $x = 7$ or $x = 8$, before making the final practical choice.